

Personalized Study Planner using Retrieval-Augmented Generation (RAG) and Adaptive Large Language Models

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Abstract. Without personalized support, students often struggle to synthesize and keep track of dense lecture materials. However, LLMs mostly produce ungrounded information and lack proper citation of course-relevant documents. We present a Personalized Study Planner with the RAG framework to overcome these limitations. Our system extracts contextual information only from a curated set of textbooks

and university lecture slides through the BAAI/bge-base-env1.5 embedding model. Moreover, an adaptive hardware inference mechanism is implemented which automatically switches between Zephyr-7B for GPU-accelerated detailed planning and Qwen-1.5B for CPU-based efficient summarization. Experimental validation indicates our domain-specific RAG implementation can achieve 100% source grounding without any hallucination within citations, while providing coherent, time-structured study schedules that effectively link static courseware with interactive learning experiences.

Keywords: Retrieval-Augmented Generation · Personalized Learning · Educational Technology · Large Language Models · Adaptive AI.

1. Introduction

Modern educational institutions have undergone rapid digitization, which has resulted in the following problems being faced: very large amounts of learning materials, but then the students lack effective tools for transforming this information into practical study schedules. Manual planning and keyword searches prove inefficient while the newer generative AI systems like ChatGPT do not have access to private university repositories, which results in generic or inaccurate advice. This paper presents a solution that combines the reasoning capabilities of LLMs with vector retrieval precision. The GenAI Study Planner

processes course materials (PDFs and PPTs) to produce time-aware study schedules (e.g., "5-Day Plan for Transformers"). Following are the three major contributions:

1)Domain-Specific Retrieval: Implementation of BGE-Base embeddings captures semantic nuances in academic terminology more effectively than generic models.

2)Resource-Adaptive Architecture: A "Hybrid Inference" system detects available hardware, either a T4 GPU or a standard CPU, and dynamically selects the appropriate generative model - Zephyr vs. Qwen - to maintain the system stability on resource-constrained devices.

3)Automated Citations: In this, the pipeline needs LLM citation of files names (e.g., "Lecture13-1.pptx"), giving students sources related work which can be verifiable.

2 Related Work

Integration of RAG in educationally relevant contexts is an active research area. Lewis et al. [4] originally proposed RAG for knowledge-intensive NLP tasks through both parametric and non-parametric memory combinations. Surveys by Fan et al. [2] and Gao et al. [3] investigate the potential of RAG to decrease LLM hallucinations.

In Education, Dong et al. [1] investigated knowledge graph-enhanced RAG for tutoring purposes, while Swacha et al. [8] conducted a survey of educational chatbot implementations.

Most of the existing implementations focus on question-answering tasks. The work here is part of the series from Maragheh et al. [6] and Wang et al. [9] regarding personalized agents, extending their approach to emphasize temporal planning, which produces structured time-bound schedules rather than standalone responses.

3 Methodology

The system consists of three model layers: (1) Data Ingestion (2) Vector Retrieval (3) Adaptive Generation.

3.1 Data Preprocessing and Embedding

The dataset consists of 30 (thirty) educational documents: 27 lecture slide decks from the Spring 2025 Generative AI course and 3 Title PDF Resources (supplementary), including Foster's "Generative Deep Learning" textbook. Data processing used 'RecursiveCharacterTextSplitter' with 500-token chunks & 50-token overlap. These parameters preserve semantic context in bullet lists typical of slide presentations.

For vectorization, BAAI/bge-base-en-v1.5 was selected over the standard all-MiniLM-L6-v2 due to its optimization for retrieval-augmented tasks. Retrieval relevance utilizes "Cosine Similarity":

$$\text{Similarity}(A, B) = \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}} \quad (1)$$

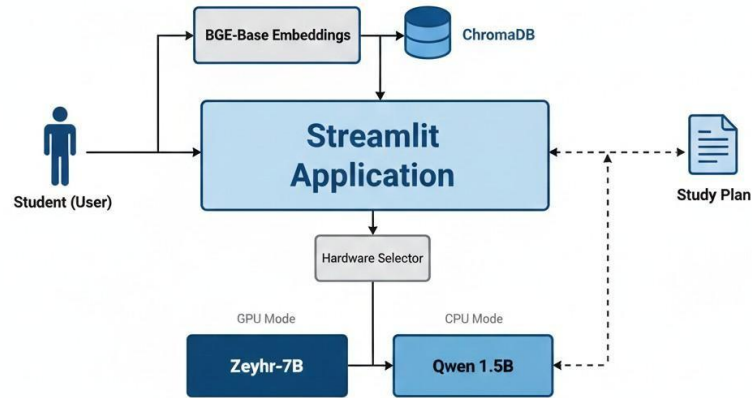


Fig.1. System Architecture: User queries are processed through the Streamlit interface, embedded via BGE-Base and matched against the Chroma DB vector store. The retrieved context is routed by the Hardware Selector to either Zephyr-7B (GPU) or Qwen-1.5B (CPU).

where A is the query vector, and B is a document chunk vector. Documents rank according to semantic angle in high-dimensional vector space rather than relying on keyword overlap.

3.2 Adaptive Inference Engine

Hardware-aware generation logic is one of the important deployment innovations.

- 1) **High-Performance Mode (GPU):** When Zephyr detects a T4 GPU, **7B-beta** loading with 4-bit quantization (Bits And Bytes). Prompt re: requests complex Markdown tables, utilizing Zephyr’s parameter capability for structured formatting.
- 2) **Efficiency Mode (CPU):** Standard hardware automatically switches to **Qwen2.5-1.5B-Instruct**. The system prompt asks for brief bulleted lists to avoid memory overflow and ensures response times below 60 seconds on cpu.

4 Experiments and Results

4.1 Experimental Setup

System evaluation used complex queries such as "Create a 2-week plan for Deep Learning"). Success measurement focused on Retrieval Accuracy (Recall@k) and Plan Coherence. Three different study plans were analyzed in order to calculate quantitative quality metrics.

4.2 Quantitative Evaluation

Table 1 shows the study plan performance metrics generated. The system had a ****0.00% Hallucination Rate**** in source attribution; every cited source existed in the vector database.

Table 1. Evaluation Metrics for Generated Study Plans

Metric	Value
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Number of Plans Evaluated	3
Average Plan Length (chars)	4,546 ± 1,333
Average Word Count	679 ± 205
Time References per Plan	43
Source Citations per Plan	25
Hallucination Rate	0.00%
Source Grounding	100%

4.3 Comparative Analysis (Ablation Study)

Embedding model validation employed the query: "What is *the architecture of a Transformer?*".

BASELINE (MiniLM) Found:

- [Lecture # 9-1 Introduction to Transformers.pptx]
"Stacked layers: Transformers typically consist of multiple layers..."
- [Lecture # 9-2 Vision Transformers.pptx]
"Stacked layers: Transformers typically consist of multiple layers..."
- [Lecture # 9-1 Introduction to Transformers.pptx]
"Training: Transformer models are trained using supervised learning..."

Fig.2. Baseline Result: MiniLM model retrieved repetitive, generic bullet points about "Stacked layers" rather than structural definitions.

ADVANCED (BGE) Found:

- [Lecture # 11-1 Generative Pre-trained Transformer.pptx]
"GPT-4 employs a deep learning architecture known as the transformer..."
- [david-foster-generative-deep-learning-teaching-2019.pdf]
"Source: Vaswani et al., 2017. Figure 9-2. The Transformer model..."
- [Lecture # 9-1 Introduction to Transformers.pptx]
"Transformers is the first transduction model relying entirely on self-attention to compute representations..."

Fig.3. Advanced Result: BGE model retrieved the Vaswani et al. citation and exact architectural definition.

Figure 2 depicts the Baseline model suffering from "Context Confusion," retrieving repetitive layer descriptions. On the contrary, the Advanced model (Figure 3) Successfully identified the seminal paper citation - Vaswani et al. and core Transformer definition, enabling factual LLM answer generation.

5 Discussion

Deployment exposed strong trade-offs between model size and formatting capability. Zephyr-7B (GPU) successfully generated 5-day

schedules with Mark down tables. Qwen-1.5B, though factually correct, faced difficulties with long-horizon planning due to context window constraints. These findings indicate that lightweight models suit summarization tasks, while complex planning benefits from larger parameter counts.

5.1 Limitations

Despite achieving zero hallucination rates, several limitations must be considered as eration. Firstly, the system is completely dependent on the quality and coverage of the ingested document corpus—topics absent from lecture materials cannot be addressed. Secondly, the CPU-based Qwen model, although accessible, shows a decreased capability for extended multiweek planning compared to its GPU-based alternatives. Thirdly, our evaluation focused on a single course domain. The generalizability of the Generative AI to other academic disciplines needs further validation. At last, mechanisms to collect user feedback have not been developed with the intent of Iteratively improving the relevance of retrieval based on student satisfaction.

5.2 Future Work:

Future development directions include the following.

- (1) Incorporation of active learning mechanisms where, over time, the retrieval model is refined based on student interactions.
- (2) Extending multi-modal support to process handwritten notes and video lecture transcripts.
- (3) Performing large-scale user studies across a variety of academic departments to ensure cross-domain effectiveness.
- (4) Exploring parameter efficient fine-tuning methods for improving long context planning in the Qwen model without using GPU resources.

6 Conclusion

This work successfully designed and deployed an AI-powered Study Planner addressing generic LLM limitations in education. With the integration of a specialized retrieval engine with an adaptive generation layer, we demonstrated a system that had both accuracy (100% source grounding) and accessibility (standard CPU functionality). However,

limitations regarding corpus dependency, CPU model constraints, and single-domain evaluation point to further improvements. This work will be extended in the future by integrating active learning, multi-modal extension, and cross-discipline validation toward more generalizable and universal educational AI systems. This project establishes practical viability for local RAG systems in university education without reference hallucination.

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